

Student Project Management System

Common Project Documentation



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1. Project Goals & Objectives

Overview

- The Student Project Management System is a role-based Web API application developed using ASP.NET Core technology. The system is designed to manage academic projects assigned to students under faculty supervision.
- The application provides separate access and permissions for Admin, Faculty, and Students. The system focuses on project allocation, task management, score tracking, progress monitoring, and secure role-based authorization.
- The system follows enterprise-level architecture practices while maintaining moderate project complexity suitable for academic implementation.

Goals

- **Design a Normalized Database Structure:**
 - Create efficient and normalized tables to manage Users, Roles, Permissions, Projects, Project Allocations, and Tasks.
 - Implement proper primary and foreign key relationships, including one-to-many and many-to-many associations wherever required to maintain data integrity and consistency.
- **Implement CRUD & Authentication Functionality:**
 - Develop Create, Read, Update, and Delete operations for all major entities such as Students, Faculty Members, Projects, Tasks, and Role Management.
 - Ensure secure user authentication and authorization using JWT tokens, along with smooth data management through role-based access control mechanisms.
- **Ensure Data Integrity and Validation:**
 - Ensure that users can easily access project information, monitor progress, and perform their assigned responsibilities based on their roles.
- **Develop an Intuitive User Interface:**
 - Provide a clean, user-friendly, and responsive frontend for efficient management of academic projects and related activities.
- **Encourage Good Software Development Practices:**
 - Apply layered architecture principles, separation of concerns, and modular coding practices to improve maintainability and scalability.
 - Utilize design patterns, DTOs, dependency injection, and proper exception handling to develop a robust and enterprise-ready application.
- **Facilitate Effective Project Monitoring and Evaluation:**
 - Enable faculty members to assign tasks, track student performance, and provide remarks and evaluations throughout the project lifecycle.
 - Automatically calculate project progress and score statistics to support informed academic decision-making.

Objectives

- **Student Management:**
 - Enable administrators to create, view, update, and manage student profiles, including personal information, contact details, academic information, and project assignments.
 - Allow students to update their own profile information and view project-related details assigned to them.
- **Faculty Management:**
 - Provide functionality to create and maintain faculty profiles, including departmental information, contact details, and areas of expertise.
 - Allow faculty members to supervise allocated projects and monitor the progress of assigned students.
- **Role and Permission Management:**
 - Enable administrators to define and manage user roles such as Admin, Faculty, and Student.
 - Implement role-based permissions to ensure that users can access only the functionalities relevant to their responsibilities.
- **Project Management:**
 - Facilitate the creation, modification, viewing, and deletion of academic projects, including project titles, descriptions, timelines, and status information.
 - Allow faculty members to initiate and manage projects under their supervision while maintaining project lifecycle records.
- **Project Allocation Management:**
 - Enable administrators to allocate projects to students and assign faculty supervisors for effective project guidance.
 - Maintain allocation records to ensure proper tracking of project ownership and faculty responsibilities.
- **Task Management:**
 - Provide functionality to create, assign, update, and manage project tasks associated with individual projects.
 - Allow faculty members to define task priorities, deadlines, and expected outcomes for students.
- **Student Performance and Score Evaluation:**
 - Facilitate the assignment of scores and remarks by faculty members based on the completion and quality of assigned tasks.
 - Maintain records of earned scores and evaluations to support academic assessment processes.
- **Progress Monitoring and Tracking:**
 - Automatically calculate and display task-wise and project-wise progress based on predefined evaluation criteria.
 - Provide students and faculty members with real-time insights into project completion status and pending activities.

- **Dashboard and Reporting:**
 - Provide role-specific dashboards displaying key information such as project statistics, task completion summaries, student performance metrics, and faculty supervision details.
 - Generate meaningful reports to assist administrators and faculty members in monitoring overall project activities.
- **Data Integrity and Validation:**
 - Ensure accurate and consistent data entry through appropriate validations, relational constraints, and business rules.
 - Prevent invalid project allocations, duplicate records, unauthorized modifications, and inconsistencies in task evaluation and progress calculations.
- **Authentication and Secure Access:**
 - Implement secure user authentication using JWT-based authorization mechanisms.
 - Restrict access to system functionalities based on user roles and permissions to maintain confidentiality and system security.

2. Scope of the Project

- **Student and Faculty Management**
 - Ability for administrators to manage student and faculty records, including personal information, contact details, and role assignments.
 - Create and maintain faculty profiles with departmental affiliations and areas of expertise.
 - Allow students to manage their profiles and access project-related information assigned to them.
- **Role and Permission Management**
 - Create, update, and manage various user roles such as Admin, Faculty, and Student.
 - Define and assign permissions to control access to different modules and functionalities within the system.
 - Ensure secure role-based access to maintain data confidentiality and system integrity.
- **Project Creation and Management**
 - Enable faculty members and administrators to create, update, view, and manage academic projects.
 - Maintain project details such as project title, description, objectives, timelines, and project status.
 - Support project lifecycle management, including project initiation, execution, completion, and archival.
- **Project Allocation and Supervision**
 - Assign students to projects under the guidance of designated faculty supervisors.
 - Maintain allocation records to track project ownership, supervision responsibilities, and allocation history.
 - Restrict faculty members to viewing and managing only the projects allocated to them.
- **Task Assignment and Progress Tracking**

- Allow faculty members to create, assign, and manage project-related tasks for individual students or project teams.
 - Define task priorities, deadlines, descriptions, and completion criteria.
 - Monitor task completion status and automatically update progress percentages based on predefined evaluation logic.
- **Input Validation and Error Handling**
 - Ensure that all project-related inputs, including project details, task assignments, allocations, and evaluations, are validated for correctness and consistency.
 - Prevent duplicate records, invalid allocations, unauthorized actions, and incorrect score calculations through business rule enforcement.
 - Provide meaningful error messages, warnings, and user-friendly notifications to guide users during system interactions.
 - **Reporting and Analytics**
 - Generate reports related to project progress, task completion, student performance, and faculty supervision activities.
 - Support data export functionalities to facilitate academic reviews and administrative reporting requirements.
 - Enable stakeholders to analyze project trends and monitor the effectiveness of project execution processes.

3. Project Timeline

Week No.	Week Date	Task List
1	29/06/2026 To 04/07/2026	Study and analysis of existing Student Project Management Systems for understanding project requirements.
2	06/07/2026 To 11/07/2026	Preparation of Common Layout for consistent UI and Different Project Screens
3	13/07/2026 To 18/07/2026	Screen Designs for List Pages and Add/Edit Pages
4	20/07/2026 To 25/07/2026	API Project creation with Model Classes and Database creation using Migrations
5	27/07/2026 To 01/08/2026	API CRUD Endpoints for 2 Modules and Consume the endpoints

6	03/08/2026 To 08/08/2026	API CRUD Endpoints for next 2 Modules and Consume the endpoints
7	10/08/2026 To 15/08/2026	API CRUD Endpoints for remaining Modules and Consume the endpoints
8	17/08/2026 To 22/08/2026	API CRUD Endpoints with Different functionalities like sorting, pagination and filtering and UI Modification as required.
9	24/08/2026 To 29/08/2026	Dashboard API Designs and Common Response for all the API endpoints and Create Dynamic UI for Dashboard.
10	31/08/2026 To 05/09/2026	Input Validations using Fluent Validations with appropriate DTOs.
11	07/09/2026 To 12/09/2026	Authentication & Authorization and Role Base Access Controls Implementation using Json Web Tokens in API Project
12	14/09/2026 To 19/09/2026	Authentication & Authorization and Role Base Access Controls Implementation using Json Web Tokens Frontend Project
13	21/09/2026 To 26/09/2026	Final Evaluation

4. Database Overview

SPM_Role		
ColumnName	DataType	Remarks
RoleId	int	Primary Key
RoleName	nvarchar(50)	Not Null
Description	nvarchar(250)	Allow Null

SPM_User		
ColumnName	DataType	Remarks
UserId	int	Primary Key
FullName	nvarchar(150)	Not Null
Email	nvarchar(150)	Not Null , Unique
Password	nvarchar(max)	Not Null

MobileNumber	nvarchar(15)	Not Null
ProfilePicturePath	nvarchar(500)	Image path, Not Null
IsActive	bit	Not Null
IsDeleted	bit	Allow Null, Soft Delete

SPM_UserRole

ColumnName	DataType	Remarks
RolePermissionId	int	Primary Key
RoleId	int	Not Null, FK → Roles
UserId	Int	Not Null, FK → Users

SPM_Status

ColumnName	DataType	Remarks
StatusID	Int	Primary Key
StatusName	Nvarchar(20)	Not Null
StatusCssClass	Nvarchar(100)	Not Null

SPM_Priority

ColumnName	DataType	Remarks
PriorityID	Int	Primary Key
PriorityName	Nvarchar(20)	Not Null
PriorityCssClass	Nvarchar(20)	Not Null

SPM_Project

ColumnName	DataType	Remarks
ProjectId	int	Primary Key
ProjectTitle	nvarchar(200)	Not Null
Description	nvarchar(max)	Allow Null
StudentId	int	Not Null, FK → Users
FacultyId	int	Not Null, FK → Users
AssignedDate	datetime	Not Null, Default GETDATE()
IsDeleted	bit	Allow Null, Soft Delete
ProjectStatus	Int	Not Null, FK → Status
StartDate	datetime	Not Null
EndDate	datetime	Not Null
TotalTasks	int	Not Null, Auto Updated
CompletedTasks	Int	Not Null, Auto Updated
ProgressPercentage	decimal(5,2)	Not Null, Auto Calculated

SPM_Task		
ColumnName	DataType	Remarks
TaskId	int	Primary Key
AllocationID	int	Not Null, FK → Allocation
TaskTitle	nvarchar(200)	Not Null
TaskDescription	nvarchar(max)	Allow Null
TaskStatus	Int	Not Null, FK → Status
PriorityID	Int	Not Null, FK → Priority
AssignedScore	decimal(5,2)	Not Null, Faculty-defined score
EarnedScore	decimal(5,2)	Allow Null, Obtained score
ProgressPercentage	decimal(5,2)	Not Null, Auto Calculated
StartDate	datetime	Allow Null
DueDate	datetime	Allow Null
CompletedDate	datetime	Allow Null
FacultyRemarks	nvarchar(500)	Allow Null
StudentRemarks	nvarchar(500)	Allow Null
IsDeleted	bit	Allow Null, Soft Delete

5. Functional Requirements

- **Role Management: Priority - 1**
 - Manage Roles
 - Prevent duplicate Roles
- **User Management: Priority - 1**
 - Manage (Create, View, Edit, Delete) Users.
 - Prevent duplicate entries
 - Maintain audit fields (Created, Modified) automatically
- **User-Role Management: Priority - 2**
 - Manage User Roles.
 - Prevent Duplicates and Assign Roles accordingly.
- **Status and Priority Management: Priority - 2**
 - Manage Status and Priority.
 - Prevent duplicate entries.
- **Project Management: Priority - 3**
 - Manage Project with Proper Faculty and Student Assignment.
 - Prevent Duplicates
- **Task Management: Priority - 3**
 - Create, update, view, and delete project tasks.
 - Assign tasks to students with priorities and deadlines.
 - Update task status (Pending, In Progress, Completed, Rejected).
 - Prevent duplicate task assignments within the same project.
 - Automatically calculate task progress percentages.

- Automatically update project completion percentages based on task evaluations.
- **Dashboard and Reporting: Priority - 4**
 - Design role-based dashboards for Admin, Faculty, and Students.
 - Provide project statistics and task completion summaries based on roles.
 - Generate reports related to student performance and project progress.
 - Support export functionality for reports in various formats where applicable.
- **Authentication and Authorization: Priority – 4**
 - Implement secure login functionality using JWT authentication.
 - Support logout and token validation mechanisms.
 - Restrict access to APIs based on assigned roles and permissions.
 - Ensure unauthorized users cannot access protected resources.

6. Screen List

1. Authentication & Dashboard

1. **Login Page:** Role-based authentication (Admin / Faculty / Student) using JWT-based authorization.
2. **Dashboard:** Displays an overview of active projects, completed projects, pending tasks, upcoming deadlines, and project statistics based on the logged-in user's role.
3. **Profile:** View and update personal information, contact details, and profile picture for the logged-in user.

2. Master Configuration

- Role Management (List Page and Add/Edit Screens)
- User Management (List, Profile, and Add/Edit Screens)
- Role-User Management (List Page and Add/Edit Screens)

3. Project and Task Management

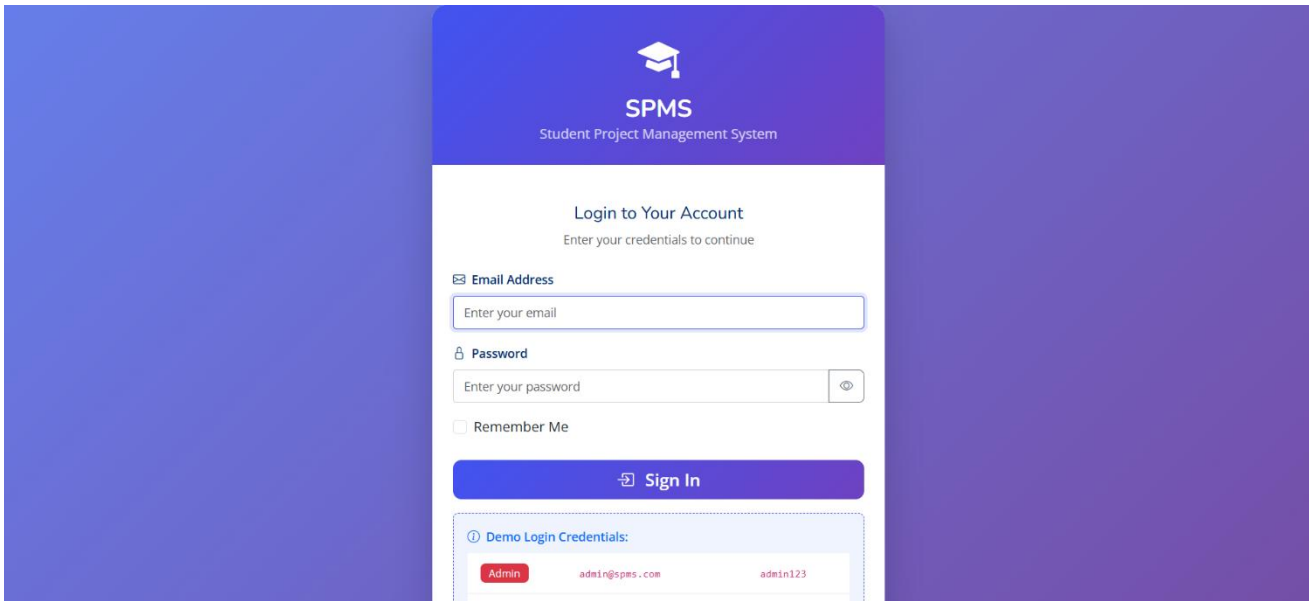
- Project Management (List Page, Detailed View and Add/Edit Screens)
- Task Management (List Page, Detailed View and Add/Edit Screens)

4. Dashboards and Reports

- Role based Dashboards (Admin/Faculty/Student Dashboards)
- Reports for Different statistics as required.

7. Screen Design Reference

Login Page



The login page features a purple header with the SPMS logo and title. Below the header, a white card contains the login form. The form includes fields for Email Address and Password, a Remember Me checkbox, and a Sign In button. Demo login credentials are provided at the bottom.

SPMS
Student Project Management System

Login to Your Account
Enter your credentials to continue

Email Address
Enter your email

Password
Enter your password

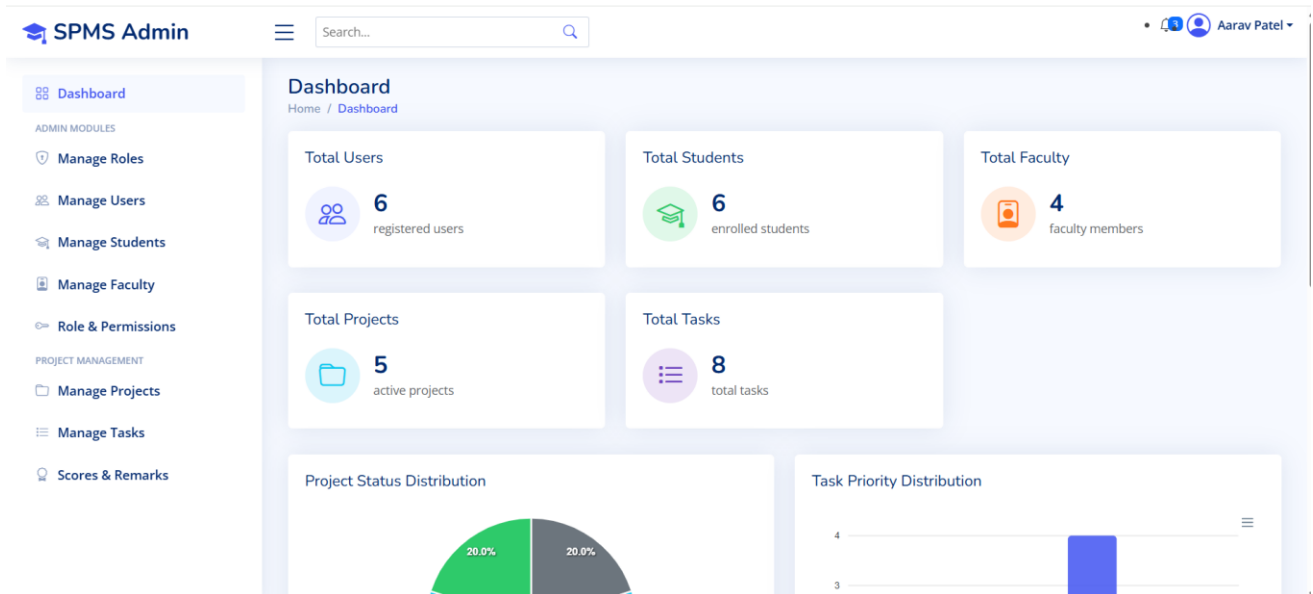
☐ Remember Me

Sign In

Demo Login Credentials:

Username	Password
Admin	admin123

Dashboard



Manage Roles

Search...

Aarav Patel

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Manage Roles

Home / Roles

Search roles...

+ Add Role

#	Role Name	Description	Actions
1	Admin	Full access to all modules and settings.	
2	Student	Can view and manage own projects and tasks.	
3	Faculty	Can supervise students and manage project assignments.	

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Designed with NiceAdmin theme

Roles Add/Edit Page

Search...

Aarav Patel

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Add Role

Home / Roles / Add

Role Details

Role Name

Enter role name

Description

Enter description

Save

Back

Manage Users

SPMS Admin

Search...

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Manage Users

Manage Users

Home / Users

Search users...

Filters

Add User

Search

Role

Status

Name, email, mobile...

All Roles

All Status

Apply

Reset

#	Full Name	Email	Mobile	Role	Status	Actions
1	Aarav Patel	admin@spms.com	9876543210	Admin	Active	Edit Delete
2	Priya Sharma	priya.sharma@spms.com	9876543211	Faculty	Active	Edit Delete
3	Rohan Mehta	rohan.mehta@spms.com	9876543212	Student	Active	Edit Delete
4	Sneha Desai	sneha.desai@spms.com	9876543213	Student	Active	Edit Delete
5	Vikram Singh	vikram.singh@spms.com	9876543214	Faculty	Inactive	Edit Delete
6	Anita Inchi	anita.inchi@nms.com	9876543215	Student	Active	Edit Delete

User Add/Edit Page

SPMS Admin

Search...

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Manage Users

Add User

Home / Users / Add

User Details

Full Name

Enter full name

Email

Enter email

Password

Enter login password

Mobile

Enter mobile number

Role

-- Select Role --

Status

Active

Save

Back

Manage Projects

SPMS Admin

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Search...

Aarav Patel

Manage Projects

Home / Projects

Search projects...

Filters

+ Add Project

Search

Status

Title, description...

All Status

Apply

Reset

#	Project Title	Status	Start Date	End Date	Assigned To	Actions
1	E-Commerce Platform Build a full-stack e-commerce web application.	In Progress	01 Sep 2024	01 Mar 2025	Priya Sharma 3 Students	✓ ✗
2	Library Management System Develop a library management system with barcode s...	Completed	15 Aug 2024	15 Feb 2025	Assigned Students Rohan Mehta Anita Joshi Arjun Reddy	✓ ✗
3	AI Chatbot Create an AI-powered chatbot for student queries.	In Progress	01 Jan 2025	30 Jun 2025	Dr. Nisha Kapoor 2 Students	✓ ✗
4	IoT Weather Station Build an IoT-based weather monitoring station.	Not Started	01 Mar 2025	01 Sep 2025		✓ ✗

Project Add/Edit Page

SPMS Admin

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Search...

Aarav Patel

Add Project

Home / Projects / Add

Project Details

Project Title

Enter project title

Description

Enter description

Start Date

13-06-2026

End Date

13-09-2026

Status

Not Started

Supervising Faculty

-- Select Faculty --

Assigned Students

Rohan Mehta
Sneha Desai
Anita Joshi
Rahul Verma

✓ Save

← Back

Manage Tasks

SPMS Admin

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Search...

Manage Tasks

Home / Tasks

Search tasks...

Filters

Add Task

Search

Priority

Status

Title, description...

All

All

Apply

Reset

#	Task Title	Project	Priority	Status	Assigned To	Due Date	Actions
1	Design Database Schema	E-Commerce Platform	High	Pending	Rohan Mehta (Student)	15 Sep 2024	Edit Delete
2	Setup Project Structure	E-Commerce Platform	Medium	Pending	Rohan Mehta (Student)	20 Sep 2024	Edit Delete
3	Implement User Authentication	E-Commerce Platform	High	In Progress	Rohan Mehta (Student)	01 Oct 2024	Edit Delete
4	Create Book Catalog Module	Library Management System	Medium	Completed	Sneha Desai (Student)	15 Oct 2024	Edit Delete
5	Train NLP Model	AI Chatbot	Critical	In Progress	Priya Sharma	01 Mar 2025	Edit Delete

Task Add/Edit

SPMS Admin

Dashboard

ADMIN MODULES

Manage Roles

Manage Users

Manage Students

Manage Faculty

Role & Permissions

PROJECT MANAGEMENT

Manage Projects

Manage Tasks

Scores & Remarks

Search...

Add Task

Home / Tasks / Add

Task Details

Task Title

Enter task title

Description

Enter description

Project

-- Select Project --

Due Date

20-06-2026

Priority

Medium

Status

Pending

Assigned Type

Student

Assigned To

-- Select Project First --

Save

Back

Best of luck!